

Climademic Suitability Model:

***Aedes aegypti* and *Aedes albopictus* Habitat Suitability**

Data Specification

Overview

This dataset includes modelled habitat suitability probabilities for the mosquito species *Aedes aegypti* and *Aedes albopictus* based on the Climademic Suitability Model. The data provide monthly estimates of environmental suitability across the globe over multiple decades from 1975 to 2024.

All details describing the used feature data sets, model implementation and analysis, and suitability discussion are presented in the accompanying manuscript “Suitable seasons: Global monthly habitat suitability for the arbovirus vectors *Aedes aegypti* and *Aedes albopictus* in 1975–2024” by Siddiqui et al.

Contact:

Christopher Irrgang, irrgangC@rki.de

Climate and Societal Analytics, Centre for Artificial Intelligence in Public Health Research,
Robert Koch Institute, Berlin, Germany.

File Format

Format: GeoTIFF (.tif)

Each file contains 12 raster layers, corresponding to the 12 months of a year (January–December).

Spatial Characteristics

Coordinate Reference System (CRS): EPSG:4326 (WGS 84)

Spatial Extent (Bounding Box):

Longitude: -180° to +180°

Latitude: +83.75° to -56°

Spatial resolution: 0.25°

Temporal Coverage

Time Range: 1975–2024

Temporal Resolution: Monthly

Data Description

Variable: Habitat suitability probability

Value Range:

Valid values: [0.0, 1.0]

0.0 → unsuitable habitat

1.0 → highly suitable habitat

No Data Value:

-1.0 indicates missing or undefined data

Data Structure

Each GeoTIFF file represents a single year of data

Each file contains:

12 bands (layers):

Band 1 → January

Band 2 → February

...

Band 12 → December

Notes

Values are probabilistic and model-derived, not direct observations

Users should account for spatial resolution and uncertainty when interpreting results

The dataset excludes regions outside the defined spatial extent or with insufficient data (marked as -1.0)

Due to data availability limitations, the suitability dataset does not include coverage for the Svalbard archipelago.